

Damage Induced Anisotropy: On the Difficulties Associated with the Active/Passive Unilateral Condition

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ABSTRACT: Continuum damage theories have been developed that incorporate both the damage induced anisotropy and the unilateral effect of damage. Four of them are discussed in this paper using respectively scalar damage variables, vectors, second-order tensors, and fourth-order damage tensors.

From the chosen examples, restricting ourselves to the case of elastic behaviour, it is demonstrated that no theory is able to reproduce simultaneously the induced anisotropy and the active/passive damage effects. In fact, the selected candidate theories either show a discontinuous stress-strain response when the unilateral condition takes place or an unacceptable nonsymmetric elastic behaviour for some loading conditions. Further studies have to be conducted in order to solve this fundamental problem.

1. INTRODUCTION

DURING THE PAST two decades, Continuum Damage Mechanics (CDM) have been greatly developed and applied in many situations. A typical school of CDM was initiated by the works of L. M. Kachanov (1958) and Rabotnov (1969) for the creep of metals. Generalizations for other loading cases have been considered and applied to various kinds of materials. Reviews of CDM can be found, for example, by Hult (1979), Chaboche (1981), and Krajcinovic (1984). Recent papers (Chaboche 1988a,b) summarize the present state of this phenomenological approach of CDM using damage effect variables in the sense of Kachanov and the effective stress/effective strain concepts.

A thermodynamic framework of such a CDM has been proposed by Chaboche (1977) in the case of the isotropic damage. Its generalization to anisotropic descriptions has been considered, which incorporates vectors [Krajcinovic and Fonseka (1981)], second-order tensors [Cordebois, Sidoroff (1979)], and fourth-order tensors [Lemaitre and Chaboche (1985), Simo and Ju (1987), Ju (1989)]. One of the difficulties in the CDM development concerns the nonlinearities associated with the unilateral nature of the damage. Roughly speaking, the damage is active only if microcracks are open. In the other case called "passive damage" by

Krajcinovic and Fonseka (1981), damage is still present (inactive damage should not be confused with a healing process) but affects the mechanical properties of the material differently.

An interesting case of application is related to the elastic materials whose characteristics evolve with the damage. This is a typical situation for concrete as well as for brittle composite materials. As a first approximation, the nonlinear behaviour of the materials and their internal changes can be associated to the damaging processes, such as voids, microcracks . . . , their initiation, growth, and coalescence. In such a case we can neglect the irreversible part of the strain or consider it as directly associated to the damage.

The purpose of the present paper is to briefly review the present capabilities of CDM theories and to describe simultaneously the anisotropic nature of damage and its unilateral character. First we shall concentrate on the macroscopic phenomenological approaches and consider some of presently known theories (Sections 3 and 4). Then we shall discuss the range of applicability and the limitations of theories based on a microscopic description of damage (Section 5). In order to simplify the presentation, we limit ourselves to the case of linear elastic damaged material under small strain, isothermal, and time-independent conditions.

2. GENERALITIES ON THE CDM DESCRIPTION OF DAMAGE

2.1 Thermodynamic Framework

As in the general case of dissipative materials described with internal variables [Germain et al. (1983)], we may consider two potentials:

1. *The thermodynamic potential*, for instance the specific free energy, shall be written in the form:

$$\Psi = \Psi(\epsilon, d_a) = \frac{1}{2} \epsilon : \underline{\Lambda} : \epsilon \quad (1)$$

$$\underline{\Lambda} = \underline{\Lambda}(d_a) \quad (2)$$

where ϵ is the elastic strain tensor and d_a denotes a family of damage variables ($a = 1, 2 \dots$). $\underline{\Lambda}(d_a)$ is a fourth-order symmetric tensor, the secant stiffness, function of damage. It corresponds here to the case of active damage (open microcracks). Linear terms in ϵ could be added to Equation (1) in order to describe the irreversible strains associated with damage [Hermann (1989), Ramtani (1990)].

From Equation (1) we define the stress and the damage energy release rate [introduced by Chaboche (1977)]:

$$\underline{\sigma} = \frac{\partial \Psi}{\partial \epsilon} = \underline{\Lambda}(d_a) : \epsilon \quad (3)$$

$$y_a = - \frac{\partial \Psi}{\partial d_a} = - \frac{1}{2} \epsilon : \frac{\partial \underline{\Lambda}}{\partial d_a} : \epsilon \quad (4)$$

2. *The dissipative potential* corresponds to the irreversible processes. In this case, it is the damage evolution. It is written in terms of the thermodynamic forces associated with damage but depends also on the present value of damage:

$$\Phi = \Phi(y_a; d_a) \quad (5)$$

In the present case, within the simplifying assumption of a time-independent process, we may replace the notion of dissipative potential and corresponding normality hypotheses by the following damage criterion:

$$g_a(y_a; d_a) = y_a - K_a(d_a) \leq 0 \quad (6)$$

Such a criterion states that the damage in the material is increased when the energy release y_a attains the damage threshold K_a and may increase it further. In fact, K_a evolves with damage with the condition $y_a = K_a(d_a)$, and stores the maximum value attained by y_a . In the criterion (6), we have assumed uncoupled damage evolutions d_a , which is sufficient for the purpose of the present paper. To be more general, coupled criteria may easily be stated instead of Equation (6).

To reflect the loading/unloading condition, evolution of damage is defined with the Kuhn-Tucker relations [see Ju (1989) for instance]:

$$\dot{d}_a \geq 0 \quad g_a(y_a; d_a) \leq 0 \quad \dot{d}_a g_a(y_a; d_a) = 0 \quad (7)$$

Within this framework, the second thermodynamic principle reduces to the following inequality:

$$\dot{\Phi}_d = y_a \dot{d}_a \geq 0 \quad (8)$$

Provided the damage energy release rate y_a is always positive, this principle is automatically checked.

In the next section, we consider several possibilities for the tensorial nature of the damage variable d_a .

2.2 The Effective Stress Concept

The notion of effective stress, first introduced by Rabotnov (1969), may be considered as follows [Chaboche (1977)]: the effective stress tensor $\tilde{\sigma}$ is a fictitious one. It is to be applied to the virgin volume element of material in order to obtain the same strain state as the one produced by the presently applied stress tensor $\underline{\sigma}$ on the damaged material. The following linear relation can be assumed between $\underline{\sigma}$ and $\tilde{\sigma}$:

$$\tilde{\sigma} = \underline{M}(d_a) : \underline{\sigma} \quad (9)$$

where $\underline{\underline{M}}$ is a fourth-order tensor that characterizes the state of damage. According to the above definition and the thermodynamic potential [Equation (1)], we easily obtain:

$$\underline{\underline{M}} = \underline{\underline{\Lambda}} : \underline{\underline{\tilde{\Lambda}}}^{-1} \quad (10)$$

where $\underline{\underline{\Lambda}}$ denotes the initial undamaged elastic stiffness.

2.3 The Tensorial Nature of Damage Variables

The simplest case relates to the description of an isotropic damage. Roughly speaking, we need only two scalar variables, say d and δ , corresponding respectively to deviatoric and volumetric effects. In several theories these two variables were considered identical, leading to:

$$\underline{\underline{\tilde{\Lambda}}} = (1 - d) \underline{\underline{\tilde{\Lambda}}} \quad \text{and} \quad \underline{\underline{\tilde{\sigma}}} = \frac{\underline{\underline{\sigma}}}{1 - d} \quad (11)$$

For the anisotropic case, two situations should be considered:

- The simplest situation corresponds to initially anisotropic material, for example, composite materials (long fibers), where the directionality of damage is entirely defined by the initial symmetries of the material. Microcracks are oriented parallel or perpendicular to the fiber directions and the initial symmetries are preserved. In such a case, it is sufficient to consider scalar variables for the description of damage in the principal axes of the (initial) material. Using such a well-defined directionality, the unilateral conditions for damage can be introduced easily. This case is not considered here.
- The difficult situation corresponds to the initially isotropic materials, where the directionality of damage has to be defined by the present state of stress (connected to its principal axes) or by its history. For a general case, the damage variable contains its own directionality (like the direction of a microcrack). Various possibilities for the tensorial nature of the damage variable are discussed below [see Krajcinovic (1984) for a more complete review].

Damage as a family of vectors. This choice corresponds well to a description of damage at a microscale, taking into account the direction of the microcracks. It was used, for example, by Krajcinovic and Fonseka (1981) and applied to the behaviour of concrete. A particular advantage is to allow a natural way for introducing the unilateral condition associated with active/passive damage. For each crack system (represented by a vector $\vec{\omega}_a$), it is sufficient to consider the sign of the conventional normal stress $\underline{\underline{\sigma}} \cdot \vec{\omega}_a \cdot \vec{\omega}_a$ (or the normal strain) as corresponding to the opening/closure condition for the cracks.

In the case of materials with a well-defined initial directionality, say composite materials, the direction of vectors $\vec{\omega}_a$ may be fixed by the microstructure, independent of the applied stresses (or strains). Such a situation was developed, for

example, by Talreja (1985). For initially isotropic materials, the direction of microcracks are given by the present stress state and its past history. For general situations with nonproportional stress paths, the directions of material space must be “meshed” in a large number of elementary directions $\vec{n}_a$, corresponding to the possible crack directions [Dienes (1989)]. This means that a large number of damage variables and some arbitrariness are at a continuum sense. Such an approach will be discussed briefly in Section 5.

Damage as second-order tensors. The geometrical properties of a given state of microcracks may be described, in a continuum sense, by a second-order tensor. This tensor defines the loss of a net section of material as in the initial approach of Kachanov (1958). It was first proposed by Vakulenko and Kachanov (1971) and used by Dragon (1976) for a study of brittle fracture of rocks and by Murakami and Ohno (1980) in a theory of anisotropic creep damage. The use of the corresponding “net stress” in the thermodynamic potential leads in general to a nonsymmetrical elastic stiffness tensor, which is not acceptable as discussed by Sidoroff (1980) and Cordebois (1983).

The second-order damage tensor $\underline{d}$ can also be defined in terms of the effective stress tensor. From symmetry considerations, Cordebois and Sidoroff (1979) use the effective stress and the associated effective strain:

$$\begin{aligned}\underline{\tilde{\sigma}} &= (\underline{1} - \underline{d})^{-1/2} \cdot \underline{\sigma} \cdot (\underline{1} - \underline{d})^{-1/2} \\ \underline{\tilde{\epsilon}} &= (\underline{1} - \underline{d})^{1/2} \cdot \underline{\epsilon} \cdot (\underline{1} - \underline{d})^{1/2}\end{aligned}\quad (12)$$

The free enthalpy and the free energy are written within the concept of energy equivalence:

$$\Psi^* = \frac{1}{2} \underline{\Lambda}^{-1} : \underline{\sigma} : \underline{\sigma} \quad (13)$$

The damage elastic compliance tensor now expresses:

$$\underline{\underline{\Lambda}}^{-1}(\underline{d}) = \underline{\underline{M}}(\underline{d}) : \underline{\underline{\Lambda}}^{-1} : \underline{\underline{M}}(\underline{d}) \quad (14)$$

where $\underline{\underline{M}}(\underline{d})$ is the fourth-order operator that defines the effective stress:

$$\underline{\tilde{\sigma}} = \underline{\underline{M}}(\underline{d})[\underline{\sigma}] \quad \underline{\tilde{\epsilon}} = \underline{\underline{M}}^{-1}(\underline{d})[\underline{\epsilon}] \quad (15)$$

This elastic stiffness tensor is symmetrical. In the principal axis for the damage

tensor (with the diagonal components d_1, d_2, d_3), $\underline{\underline{\Lambda}}(\underline{\underline{d}})$ expresses for the initially isotropic material using Voigt notations:

$$\underline{\underline{\Lambda}}^{-1}(\underline{\underline{d}}) = \begin{bmatrix} A_1 & B_{12} & B_{13} & & & \\ B_{12} & A_2 & B_{23} & & & \\ B_{13} & B_{23} & A_3 & & & \\ & & & C_{23} & & \\ & & & & C_{31} & \\ & & & & & C_{12} \end{bmatrix}$$

$$A_i = \frac{1}{E(1 - d_i)^2} \quad B_{ij} = -\frac{\nu}{E(1 - d_i)(1 - d_j)}$$

$$C_{ij} = \frac{1 + \nu}{E(1 - d_i)(1 - d_j)} \quad (16)$$

Such a theory offers interesting possibilities and the serious advantage of describing the actual state of damage by only a symmetric second-order tensor. It was also developed and generalized by Lee et al. (1985) and by Chow and Wang (1987).

However, the damage elastic stiffness tensor and the damage energy release rate are difficult to express in the general case. Moreover, the physical meaning of the energy release rate is not clear and some inconsistencies do appear when using it in the expression for the damage loading surface [see Chow and Wang (1987)].

Damage as a fourth-order tensor. This is the most natural way to introduce the effective stress and effective behaviour of the elastic material without the above-mentioned restrictions. It was proposed by Chaboche (1979) on the basis of classical homogenization results for the cracked elastic material. It can be easily shown that a fourth-order non-symmetrical operator relates the initial and present elastic stiffnesses. This operator was considered as the damage tensor:

$$\underline{\underline{\tilde{\Lambda}}} = (\underline{\underline{1}} - \underline{\underline{D}}) : \underline{\underline{\Lambda}} \quad (17)$$

where $\underline{\underline{1}}$ is the identity fourth-order tensor. In that case, we can measure the present damage tensor from the measures of the actual stiffness:

$$\underline{\underline{D}} = \underline{\underline{1}} - \underline{\underline{\tilde{\Lambda}}} : \underline{\underline{\Lambda}}^{-1} \quad (18)$$

The effective stress and the damage energy release rate express easily:

$$\underline{\underline{\sigma}} = (\underline{\underline{1}} - \underline{\underline{D}})^{-1} : \underline{\underline{\sigma}} \quad (19)$$

$$\underline{\underline{Y}} = \frac{1}{2}(\underline{\underline{\Lambda}} : \underline{\underline{\epsilon}}) \otimes \underline{\underline{\epsilon}} = \frac{1}{2} \underline{\underline{\sigma}} \otimes \underline{\underline{\epsilon}} \quad (20)$$

where $\otimes$ denotes the tensorial product.

A phenomenological damage rate equation was proposed in the case of creep of metals, with the simplifying assumption for damage rate to be proportional to a constant fourth-order tensor given in the principal axes of the effective stress. This tensor was taken as a linear combination:

$$\dot{\underline{\underline{D}}} = \underline{\underline{Q}} \dot{D} = [\gamma \underline{\underline{1}} + (1 - \gamma) \underline{\underline{G}}] \dot{D} \quad (21)$$

where $\underline{\underline{G}}$ was defined from the analysis of an elastic solid containing parallel microcracks. $\dot{D}$ defines the rate of a scalar damage variable and contains the nonlinearity of the damage evolution [see Chaboche (1979)].

More recently, other fourth-order damage tensors have been used, for instance, in ceramic-ceramic composites [Baste-Gerard (1989)] or for concrete [Simo and Ju (1987)]. In the last approach, the internal variable is taken as the current damage stiffness $\underline{\underline{\Lambda}}$ itself. This theory will be considered in Section 3.3, including the unilateral condition proposed by Ju (1989).

3. THREE TYPICAL CDM THEORIES

In this section, we present three typical CDM theories, having the capabilities to describe:

- the difference between the damaging processes under tensile and compressive loading types
- the change of the elastic stiffness when the principal stresses (or strains) change sign (the unilateral damage effect)

They are selected in order to include three descriptions with different natures for the damage variable (scalar, second-order and fourth-order tensors).

3.1 Unilateral Damage Model with Scalar Variables

Such a theory was proposed by Ladeveze and Lemaitre (1984), Mazars (1985) and summarized by Mazars and Pijaudier-Cabot (1989). The material is elastic-brittle with two scalar damage variables, D_t and D_c , respectively, for tension and compression stresses. To introduce the unilateral condition (in stresses), one uses a stress decomposition into a positive $\underline{\underline{\sigma}}_+$ and a negative $\underline{\underline{\sigma}}_-$ part. In the principal stress system, we write:

$$\underline{\underline{\sigma}}_+ = \begin{bmatrix} \langle \sigma_1 \rangle & & \\ & \langle \sigma_2 \rangle & \\ & & \langle \sigma_3 \rangle \end{bmatrix} \quad \underline{\underline{\sigma}}_- = - \begin{bmatrix} \langle -\sigma_1 \rangle & & \\ & \langle -\sigma_2 \rangle & \\ & & \langle -\sigma_3 \rangle \end{bmatrix} \quad (22)$$

where the Mac-Cauley bracket is defined by $\langle x \rangle = x$ if $x \geq 0$, $\langle x \rangle = 0$ if $x < 0$. Clearly we have $\underline{\underline{\sigma}} = \underline{\underline{\sigma}}_+ + \underline{\underline{\sigma}}_-$. We also define:

$$(\text{Tr } \underline{\underline{\sigma}})_+ = \langle \text{Tr } \underline{\underline{\sigma}} \rangle \quad (\text{Tr } \underline{\underline{\sigma}})_- = -\langle -\text{Tr } \underline{\underline{\sigma}} \rangle \quad (23)$$

Such a decomposition is formally identical with the one used for strain by Ortiz (1985), Simo and Ju (1987), that is presented in Section 3.3.

The model is defined by the thermodynamic potential (present elastic behaviour) and by a damage loading surface (governing the damage evolution). The material is considered initially isotropic and the free enthalpy is written:

$$\Psi^* = \frac{1}{2} \left\{ \frac{1}{E(1-D_t)} [(1+\nu)\underline{\sigma}_+ : \underline{\sigma}_+ - \nu(\text{Tr } \underline{\sigma})^2] + \frac{1}{E(1-D_c)} [(1+\nu)\underline{\sigma}_- : \underline{\sigma}_- - \nu(\text{Tr } \underline{\sigma})^2] \right\} \quad (24)$$

The present elastic behaviour is obtained by:

$$\underline{\varepsilon} = \frac{\partial \Psi^*}{\partial \underline{\sigma}} = \frac{1}{E(1-D_t)} [(1+\nu)\underline{\sigma}_+ - \nu(\text{Tr } \underline{\sigma})_+ \underline{1}] + \frac{1}{E(1-D_c)} [(1+\nu)\underline{\sigma}_- - \nu(\text{Tr } \underline{\sigma})_- \underline{1}] \quad (25)$$

where $\underline{1}$ is the identity second-order tensor.

The two damage energy release rates, corresponding to D_t and D_c , appear as scalars:

$$Y_t = \frac{\partial \Psi^*}{\partial D_t} \quad Y_c = \frac{\partial \Psi^*}{\partial D_c} \quad (26)$$

We can check that Y_t and Y_c are always positive; therefore, the second thermodynamic principle is always verified:

$$\Phi_d = Y_t \dot{D}_t + Y_c \dot{D}_c \geq 0 \quad (27)$$

The damage loading surface is expressed as a double-criterion [see also Equation (6)]:

$$\begin{aligned} g_t(\underline{\varepsilon}, D_t) &= Y_t - K_t(D_t) \leq 0 \\ g_c(\underline{\varepsilon}, D_c) &= Y_c - K_c(D_c) \leq 0 \end{aligned} \quad (28)$$

K_t and K_c are the hardening-softening parameters and will store the maximum values attained respectively by Y_t and Y_c .

3.2 Unilateral Damage Model with Second-Order Damage Tensors

In order to illustrate the case with a second-order tensorial damage, we take the recent theory proposed by Ramtani (1990). In this theory, the possibility for ir-

reversible strains associated with damage is considered without introducing any additional (inelastic) strain (without the corresponding dissipation). Our purpose here is not related to this problem; therefore, we do not consider the corresponding terms in the free energy:

$$\begin{aligned}\Psi = & \mu[(\underline{1} - \underline{d})^{1/2} \cdot \langle \underline{\epsilon} \rangle_+ : [(\underline{1} - \underline{d})^{1/2} \cdot \langle \underline{\epsilon} \rangle_+] \\ & + \mu[(\underline{1} - \underline{d}^*)^{1/2} \cdot \langle \underline{\epsilon} \rangle_- : [(\underline{1} - \underline{d}^*)^{1/2} \cdot \langle \underline{\epsilon} \rangle_-] \\ & + \frac{1}{2} \lambda(1 - \delta)(\text{Tr } \underline{\epsilon})^2\end{aligned}\quad (29)$$

The material is initially isotropic with the Lamé's coefficients λ and μ . $\underline{d}$ is the second-order symmetric damage tensor, $\underline{1}$ the second-order identity tensor. The "positive" and "negative" strains $\langle \underline{\epsilon} \rangle_+$ and $\langle \underline{\epsilon} \rangle_-$ are defined in the same way as above for the stresses. Let us note that the same decomposition is used also by Ortiz (1985) and Ju (1989) with a different presentation (see Section 3.3). The damage plays the role linearly through the following definition for the tensor: $\underline{x}^{1/2} \cdot \underline{x}^{1/2} = \underline{x}$.

Here, $\underline{d}^*$ plays the role mainly under compressive stresses. It is called "transmitted" damage and corresponds to microcracks parallel to the compressive stress direction. $\underline{d}$ is called "direct" damage and represents cracks perpendicular to the maximum tensile stress. The scalar damage δ introduces volumetric damaging effects.

Hooke's law for the damaged material is obtained by:

$$\begin{aligned}\underline{\sigma} = & 2\mu(\underline{1} - \underline{d})^{1/2} \cdot \langle \underline{\epsilon} \rangle_+ \cdot (\underline{1} - \underline{d})^{1/2} \\ & + 2\mu(\underline{1} - \underline{d}^*)^{1/2} \cdot \langle \underline{\epsilon} \rangle_- \cdot (\underline{1} - \underline{d}^*)^{1/2} + \lambda(1 - \delta)(\text{Tr } \underline{\epsilon}) \underline{1}\end{aligned}\quad (30)$$

The thermodynamic forces associated to damage variables are the symmetric second-order tensors Y and Y^* and the scalar Y_δ . They are difficult to express because of the derivatives of $(\underline{1} - \underline{d})^{1/2}$.

For this reason, Ramtani (1990) uses three independent damage loading surfaces expressed directly from the strain:

$$\begin{aligned}g &= \frac{1}{2} \langle \underline{\epsilon} \rangle_+ : \langle \underline{\epsilon} \rangle_+ - K(\underline{d}) \leq 0 \\ g^* &= \frac{1}{2} \langle \underline{\epsilon} \rangle_- : \langle \underline{\epsilon} \rangle_- - K^*(\underline{d}^*) \leq 0 \\ g_\delta &= \frac{1}{2} (\text{Tr } \underline{\epsilon})^2 - K_\delta(\delta) \leq 0\end{aligned}\quad (31)$$

These three damage surfaces can be activated independently or simultaneously, depending on the considered straining conditions.

3.3 The Anisotropic Damage Theory Developed by Ju (1989)

The state variable corresponding to the damaging processes is a fourth-order tensor. The special choice, made by Simo and Ju (1987), is to consider the secant elastic modulus itself as the internal variable. In order to distinguish from the situation of Section 2.3, we call it $\underline{\underline{C}}$. Its initial value is denoted $\underline{\underline{C}}_0$. The free energy is:

$$\Psi = \frac{1}{2} \underline{\underline{C}} : \underline{\underline{\epsilon}} : \underline{\underline{\epsilon}} \quad (32)$$

The thermodynamic forces are:

$$\underline{\underline{\sigma}} = \underline{\underline{C}} : \underline{\underline{\epsilon}} \quad \text{and} \quad \underline{\underline{Y}} = \frac{\partial \Psi}{\partial \underline{\underline{C}}} = \frac{1}{2} \underline{\underline{\epsilon}} \otimes \underline{\underline{\epsilon}} \quad (33)$$

where the elastic energy release is given by the fourth-order tensor $\underline{\underline{Y}}$.

In this theory the microcrack opening/closing unilateral effect is introduced by considering the spectral decomposition of the strain tensor [see also Ortiz (1985)]:

$$\underline{\underline{\epsilon}} = \sum_{i=1}^3 \epsilon_i \bar{\underline{\underline{p}}}_i \otimes \bar{\underline{\underline{p}}}_i \quad |\bar{\underline{\underline{p}}}_i| = 1 \quad (34)$$

where ϵ_i is the i -th principal strain and $\bar{\underline{\underline{p}}}_i$ is the corresponding unit principal direction. Let $\underline{\underline{Q}}$ and $\underline{\underline{Q}}^+$ respectively be the regular and positive spectral projection tensors:

$$\underline{\underline{Q}} = \sum_{i=1}^3 \bar{\underline{\underline{p}}}_i \otimes \bar{\underline{\underline{p}}}_i \quad \underline{\underline{Q}}^+ = \sum_{i=1}^3 H(\epsilon_i) \bar{\underline{\underline{p}}}_i \otimes \bar{\underline{\underline{p}}}_i \quad (35)$$

where $H(\cdot)$ is the Heaviside function. The fourth-order positive projection tensor $\underline{\underline{P}}^+$ is now introduced with components:

$$P_{ijkl}^+ = Q_{ia}^+ Q_{jb}^+ Q_{ka} Q_{lb} \quad (36)$$

So that $\underline{\underline{\epsilon}}^+$ can be expressed as:

$$\underline{\underline{\epsilon}}^+ = \underline{\underline{P}}^+ : \underline{\underline{\epsilon}} \quad (37)$$

Let us note the complete identity between the present definition of $\underline{\underline{\epsilon}}^+$ and the definition used in the previous section.

With this notation at hand, Ju (1989) defines successively the “total stiffness loss tensor”:

$$\underline{\underline{C}}^d = \underline{\underline{C}}_o - \underline{\underline{C}} \quad (38)$$

the “active stiffness loss tensor” $\underline{\underline{C}}_{\text{act}}^d$ and the “active damaged secant stiffness tensor” $\underline{\underline{C}}_{\text{act}}$ as follows:

$$\underline{\underline{C}}_{\text{act}}^d = \underline{\underline{P}}^+ : \underline{\underline{C}}^d : \underline{\underline{P}}^+ \quad \underline{\underline{C}}_{\text{act}} = \underline{\underline{C}}_o - \underline{\underline{C}}_{\text{act}}^d \quad (39)$$

The stress/strain relationship then takes the form:

$$\underline{\underline{\sigma}} = \underline{\underline{C}}_{\text{act}} : \underline{\underline{\epsilon}} \quad (40)$$

$$\underline{\underline{C}}_{\text{act}} = \underline{\underline{C}}_o - \underline{\underline{P}}^+ : (\underline{\underline{C}}_o - \underline{\underline{C}}) : \underline{\underline{P}}^+ \quad (41)$$

Formally, the present active stiffness corresponds to the particular cases $D_e = 0$ and $\underline{\underline{d}}^* = 0$ in the two previous theories.

The damage loading surface is defined by the function:

$$g = G(\underline{\underline{Y}}) - r \leq 0 \quad (42)$$

where r depends on the loading history. The damage process is then characterized by the following irreversible dissipative equations of evolution:

$$\dot{\underline{\underline{C}}} = -\dot{\mu} \frac{\partial g}{\partial \underline{\underline{Y}}} \quad (43)$$

$$\dot{\mu} \geq 0 \quad g \leq 0 \quad \dot{\mu} g \equiv 0 \quad (44)$$

For a given strain history, the actual damaged elastic modulus $\underline{\underline{C}}$ renders maximum damage energy dissipation.

To properly include the tensor $\underline{\underline{Y}}$ into function G , Ju (1989) defines a characteristic damage measure ξ such that:

$$\frac{\partial \xi}{\partial \underline{\underline{Y}}} = \underline{\underline{P}}^+ : \underline{\underline{C}}_o : \underline{\underline{P}}^+ \quad (45)$$

which leads to:

$$\xi = \frac{1}{2} \underline{\underline{\epsilon}}^+ : \underline{\underline{C}}_o : \underline{\underline{\epsilon}}^+ \quad (46)$$

Then Equation (42) is rewritten as follows:

$$g \equiv \bar{G}(\xi) - r \leq 0 \quad (47)$$

Denoting $H = \partial \bar{G} / \partial \xi$ and $\dot{r} \equiv \dot{\mu} H$, we observe that $\dot{r} = \dot{\xi} H$ in the event of damage loading. Hence, we obtain $\dot{\mu} = \dot{\xi}$ and the anisotropic damage evolution Equation (43) then becomes:

$$\dot{\underline{\underline{C}}} = -\dot{\xi} H \underline{\underline{P}}^+ : \underline{\underline{C}}_o : \underline{\underline{P}}^+ \quad (48)$$

The Second Principle is always satisfied as it is easily verified from:

$$\Phi_d = -\text{Tr}(\underline{\underline{Y}} : \dot{\underline{\underline{C}}}) = \frac{1}{2} \dot{\xi} H \underline{\underline{\epsilon}}^+ : \underline{\underline{C}}_o : \underline{\underline{\epsilon}}^+ \geq 0 \quad (49)$$

4. COMPARATIVE STUDY OF THE THREE THEORIES

4.1 The Considered Properties for the Models

In every case we have considered an initially isotropic elastic material. Among various aspects we want to find a theory with the following capabilities and properties:

1. Anisotropy of damage in the following sense: after a prior damaging history, the actual elastic behaviour of the material is anisotropic, especially under tensile conditions. The formulation must be general; it has to be applicable for varying principal strain (or stress) directions.
2. The active/passive unilateral damage condition must be introduced in the elastic behaviour. Moreover, it is necessary to distinguish two regimes for damage growth, under tensile and compressive types of loading.
3. The active/passive secant stiffness must be symmetrical (Onsager Principle):

$$C_{ijkl}^{\text{act}} = C_{klij}^{\text{act}}$$

4. The damage loading surfaces must be continuous and preferably convex in its space of definition (for instance strain space or stress space).
5. The stress (or strain) evolution is requested to be continuous when the unilateral condition takes place.

In the next section, the possibilities of the three previously mentioned theories are discussed on the basis of the above five points. In order to check the properties, we consider the particular cases of plane stress (or plane strain) written in the corresponding principal system. Continuity of the unilateral condition is observed when the stress σ_1 (strain ϵ_1) changes from a positive to a negative value, the stress σ_2 being positive.

4.2 Model with Scalar Variables

Clearly, the model of Section 3.1 is not able to reproduce the anisotropy induced by damage. However, it is simple to understand and to use. It describes the unilateral aspect of damage and the distinct damage regimes for tensile and com-

pressive loadings. In the particular case of $\sigma_3 = 0$ and in the principal stress system, we can write Hooke's law considering four regions:

$$\left. \begin{array}{l} \sigma_1 > 0 \\ \sigma_2 > 0 \end{array} \right\} \begin{cases} \epsilon_1 = \frac{1}{E(1 - D_t)} (\sigma_1 - \nu \sigma_2) \\ \epsilon_2 = \frac{1}{E(1 - D_t)} (-\nu \sigma_1 + \sigma_2) \end{cases} \quad (50)$$

$$\left. \begin{array}{l} \sigma_1 < 0 \\ \sigma_2 > 0 \\ \sigma_1 + \sigma_2 > 0 \end{array} \right\} \begin{cases} \epsilon_1 = \left[\frac{1 + \nu}{E(1 - D_c)} - \frac{\nu}{E(1 - D_t)} \right] \sigma_1 - \frac{\nu}{E(1 - D_t)} \sigma_2 \\ \epsilon_2 = -\frac{\nu}{E(1 - D_t)} \sigma_1 + \frac{\sigma_2}{E(1 - D_t)} \end{cases} \quad (51)$$

$$\left. \begin{array}{l} \sigma_1 < 0 \\ \sigma_2 > 0 \\ \sigma_1 + \sigma_2 < 0 \end{array} \right\} \begin{cases} \epsilon_1 = \frac{\sigma_1}{E(1 - D_c)} - \frac{\nu}{E(1 - D_c)} \sigma_2 \\ \epsilon_2 = -\frac{\nu}{E(1 - D_c)} \sigma_1 + \left[\frac{1 + \nu}{E(1 - D_t)} - \frac{\nu}{E(1 - D_c)} \right] \sigma_2 \end{cases} \quad (52)$$

$$\left. \begin{array}{l} \sigma_1 < 0 \\ \sigma_2 < 0 \end{array} \right\} \begin{cases} \epsilon_1 = \frac{1}{E(1 - D_c)} (\sigma_1 - \nu \sigma_2) \\ \epsilon_2 = \frac{1}{E(1 - D_c)} (-\nu \sigma_1 + \sigma_2) \end{cases} \quad (53)$$

Let us note that $\sigma_{12} = 0$ induces immediately $\epsilon_{12} = 0$.

The symmetry of the damaged compliance tensor can be checked in every case, especially for the two intermediate cases (the relation between ϵ_1 and σ_2 compared to the relation between ϵ_2 and σ_1). Similarly, the continuity of the strain response is preserved for any stress evolution. For example, for a transition from region 1 to region 2, with $\sigma_1 = 0$ and $\sigma_2 \neq 0$, Equations (50) and (51) give the same results for ϵ_1 and ϵ_2 .

By considering the values of Y_t and Y_c , given by Equation (26) in the four regions, we are able to check the continuity of the damage loading surfaces (see Figure 1), but we observe its non-convexity in the stress space.

4.3 Model with Second-Order Tensors

We consider an initial damaging process that induces a given damage state described by the two tensors $\underline{d}$ and $\underline{d}^*$. Equation (30) is rewritten with simplified notations:

$$\underline{\sigma} = \underline{h} \cdot \langle \underline{\epsilon} \rangle_+ \cdot \underline{h} + \underline{h}^* \cdot \langle \underline{\epsilon} \rangle_- \cdot \underline{h}^* + \lambda(1 - \delta)(\text{Tr } \underline{\epsilon}) \underline{1} \quad (54)$$

where the second-order symmetric tensor $\underline{h}$ is defined by $\underline{h} = \sqrt{2\mu}(1 - \underline{d})^{1/2}$.

The present loading conditions are assumed to be noncoaxial with the ones that have created the present damage state. We consider only the plane strain components ($\epsilon_3 = 0$) and write Hooke's law in the actual principal strain system. In that case, the damage effect is nondiagonal and the stress components are in two regions:

$$\begin{matrix} \epsilon_1 > 0 \\ \epsilon_2 > 0 \end{matrix} \left\{ \begin{array}{l} \sigma_{11} = [h_{11}^2 + \lambda(1 - \delta)]\epsilon_1 + [h_{12}^2 + \lambda(1 - \delta)]\epsilon_2 \\ \sigma_{22} = [h_{12}^2 + \lambda(1 - \delta)]\epsilon_1 + [h_{22}^2 + \lambda(1 - \delta)]\epsilon_2 \\ \sigma_{12} = h_{12}h_{11}\epsilon_1 + h_{12}h_{22}\epsilon_2 \end{array} \right. \quad (55)$$

$$\begin{matrix} \epsilon_1 < 0 \\ \epsilon_2 > 0 \end{matrix} \left\{ \begin{array}{l} \sigma_{11} = [h_{11}^{*2} + \lambda(1 - \delta)]\epsilon_1 + [h_{12}^2 + \lambda(1 - \delta)]\epsilon_2 \\ \sigma_{22} = [h_{12}^{*2} + \lambda(1 - \delta)]\epsilon_1 + [h_{22}^2 + \lambda(1 - \delta)]\epsilon_2 \\ \sigma_{12} = h_{12}^*h_{11}^*\epsilon_1 + h_{12}h_{22}\epsilon_2 \end{array} \right. \quad (56)$$

In the third region ($\epsilon_1 < 0, \epsilon_2 < 0$), we have only to replace h by h^* in Equation (55).

The present theory is able to reproduce the anisotropy of damage, as shown, for instance, by the nonzero shear stress under a zero shear strain (for nonpropor-

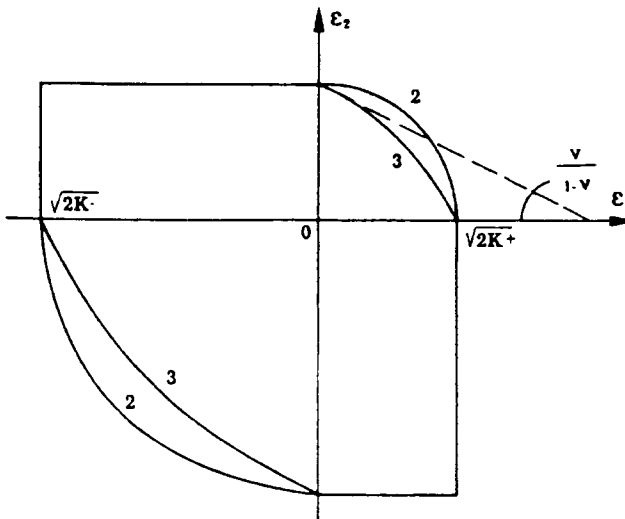


Figure 1. The loading surface in the principal stress space ($\sigma_3 = 0$) for the theory with two scalar damage variables: $K_t' = 2E(1 - D_t)K_t$, $K_c' = 2E(1 - D_c)K_c$.

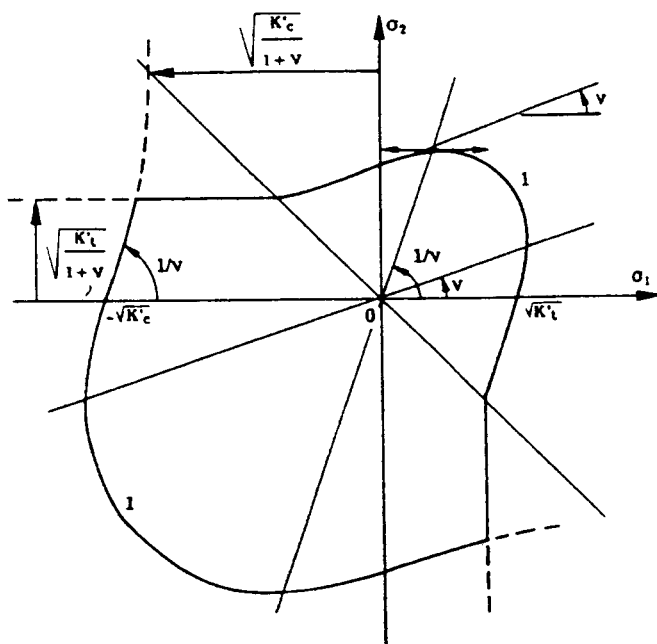


Figure 2. The loading surfaces in the principal strain space ($\epsilon_3 = 0$) for (2) the theory with two second-order tensors (the K_0 surface is not represented) and (3) the theory with one fourth-order tensor. $K^\pm = [(1 + \nu)(1 - 2\nu)/E(1 - \nu)]r^\pm$, $G(\xi) = \xi$.

tional loadings). Damage grows differently for tensile and compressive-type loadings, thanks to the different criteria [Equation (31)]. Moreover, the damage loading surface is continuous and convex as illustrated in Figure 2, and the continuity of stress response is preserved when the unilateral condition takes place. This can be checked for ($\epsilon_1 = 0$, $\epsilon_2 \neq 0$) by comparing Equations (55) and (56).

However, this theory suffers a great inconsistency: the active secant stiffness tensor is not symmetrical! In the case with $\epsilon_1 < 0$, $\epsilon_2 > 0$, Equation (56) shows different factors h_{12}^* and h_{12}^{*2} in the relations between ϵ_2, σ_{11} and between ϵ_1, σ_{22} .

4.4 The Anisotropic Damage Theory of Ju (1989)

The theory is able to reproduce a damage induced anisotropy in cases where the initial damage is created under mixed strains ($\epsilon_1 > 0$, $\epsilon_2 < 0$, $\epsilon_3 = 0$, for example). However, in the particular case where all principal strains are positive (or zero), the fourth-order positive projection tensor $\underline{P}^+$ reduces to identity and the evolution equation for damage [Equation (48)] reduces to:

$$\dot{\underline{\underline{C}}} = -\dot{\xi} H \underline{\underline{C}}_0 \quad (57)$$

so that the initial isotropy in $\underline{C}_o$ is preserved in $\underline{C}$ after damage. In fact, the induced anisotropy in this model is insufficient.

In order to study the continuity properties of the model, we restrict ourselves to the particular case where the initial damage is produced under the above conditions. Then $\underline{C}_o - \underline{C}$ is proportional to $\underline{C}_o$ and we write:

$$\underline{C} = (1 - d)\underline{C}_o \quad (58)$$

where d is a scalar damage. Considering plane strains, we can write the present stiffness tensor as:

$$\underline{C} = (1 - d) \begin{bmatrix} \lambda + 2\mu & \lambda & 0 \\ \lambda & \lambda + 2\mu & 0 \\ 0 & 0 & 2\mu \end{bmatrix} \quad (59)$$

In the principal strain system ($\epsilon_3 = 0$), Hooke's law can be written within three regions where the active stiffness is given by Equation (41):

$$\begin{array}{l} \epsilon_1 > 0 \\ \epsilon_2 > 0 \end{array} \left| \begin{array}{l} \sigma_{11} = (1 - d)(\lambda + 2\mu)\epsilon_1 + (1 - d)\lambda\epsilon_2 \\ \sigma_{22} = (1 - d)\lambda\epsilon_1 + (1 - d)(\lambda + 2\mu)\epsilon_2 \end{array} \right. \quad (60)$$

$$\begin{array}{l} \epsilon_1 < 0 \\ \epsilon_2 > 0 \end{array} \left| \begin{array}{l} \sigma_{11} = (\lambda + 2\mu)\epsilon_1 + \lambda\epsilon_2 \\ \sigma_{22} = \lambda\epsilon_1 + (1 - d)(\lambda + 2\mu)\epsilon_2 \end{array} \right. \quad (61)$$

$$\begin{array}{l} \epsilon_1 < 0 \\ \epsilon_2 < 0 \end{array} \left| \begin{array}{l} \sigma_{11} = (\lambda + 2\mu)\epsilon_1 + \lambda\epsilon_2 \\ \sigma_{22} = \lambda\epsilon_1 + (\lambda + 2\mu)\epsilon_2 \end{array} \right. \quad (62)$$

In the first region, the positive projection tensor reduces to the identity, in the third region it was zero, and in the second region it was obtained as:

$$\underline{P}^* = \begin{bmatrix} 0 & & \\ & 1 & \\ & & 0 \end{bmatrix} \quad (63)$$

The symmetry of $\underline{C}_{act}$ is checked in every case. However, we observe a discontinuity of the stress response between regions 1 and 2 when $\epsilon_1 = 0$ and $\epsilon_2 \neq 0$. This fact would lead to unacceptable results for complex multiaxial loading conditions.

The damage loading surface is continuous and convex. Its shape in the principal strain space is illustrated in Figure 2. A second damage criterion can be introduced for compressive loadings:

$$g^- = G^-(\xi^-) - r^- \leq 0 \quad (64)$$

so that we can write the damage evolution as Ju (1989):

$$\dot{\underline{\underline{C}}} = -\dot{\xi}^+ H^+ \underline{\underline{P}}^+ : \underline{\underline{C}}_o : \underline{\underline{P}}^+ - \dot{\xi}^- H^- \underline{\underline{P}}^- : \underline{\underline{C}}_o : \underline{\underline{P}}^- \quad (65)$$

5. THEORIES BASED ON A MICROCRACK DESCRIPTION

The theories studied above were developed through a direct macroscopic description of damage. Several limitations were discussed, especially in the case of the anisotropic and unilateral damage. In fact, we shall see in the present section that theories based on a microcrack description also suffer some deficiencies. We consider successively the models developed by Krajcinovic and Fonseka (1981), which describe damage through families of vectors and some proposals made, for instance, by Andrieux, Bamberger, and Marigo (1986). Here, we only consider the mechanical response of a previously damaged material. The formulation of the damage surface and of the damage growth are not discussed in the present section.

5.1 The Theory of Krajcinovic and Fonseka (1981)

These authors have proposed to describe damage by families of vectors, each of them associated with sets of (parallel) microcracks:

$$\{\vec{\omega}^a\}_{a=1;2\dots n}$$

Each vector $\vec{\omega}^a$ can be written as $\omega^a \vec{n}^a$, where $\vec{n}^a$ is the unit normal to the microcracks and ω^a is the microcrack area per unit volume.

The theory has been formulated within a general thermodynamic framework. In the particular case of the elastic-damaged material, the free energy is expressed as follows, with a combination of all the active crack systems:

$$\Psi = \frac{1}{2} \Lambda_{ijkl} \epsilon_{ij} \epsilon_{kl} + \sum_{\alpha} \sum_{\beta} [C_1 \epsilon_{kl} \epsilon_{mm} \omega_k^{\alpha} \omega_l^{\beta} + C_2 \epsilon_{kl} \epsilon_{lm} \omega_k^{\alpha} \omega_m^{\beta}] \quad (66)$$

where Λ_{ijkl} is the initial elastic stiffness of the material, and C_1 , C_2 are two material coefficients.

Using the Relation (3), we can express the stress in the form:

$$\sigma_{ij} = \frac{\partial \Psi}{\partial \epsilon_{ij}} = \tilde{\Lambda}_{ijkl} \epsilon_{kl} \quad (67)$$

where the elastic behaviour of the damaged material is given as (considering the symmetry of the strain tensor):

$$\tilde{\Lambda}_{ijkl} = \Lambda_{ijkl} + \sum_{\alpha} \sum_{\beta} [C_1(\delta_{ij}\omega_k^{\alpha}\omega_l^{\beta} + \delta_{kl}\omega_i^{\alpha}\omega_j^{\beta}) + C_2(\delta_{ik}\omega_j^{\alpha}\omega_l^{\beta} + \delta_{jl}\omega_i^{\alpha}\omega_k^{\beta})] \quad (68)$$

The above expression of elastic stiffness of the damaged material stands for active damage conditions, that is, among the systems $a = 1, 2, \dots, n$, only the ones corresponding to a positive normal strain are retained. More precisely, for family a

$$\vec{\omega}^a = \omega_1^a \vec{e}_1 + \omega_2^a \vec{e}_2 + \omega_3^a \vec{e}_3$$

where $\vec{e}_1, \vec{e}_2, \vec{e}_3$ is the chosen coordinate system, we use:

$$\begin{cases} \omega_i^a \neq 0 & \text{if } \epsilon_a = \omega_k^a \epsilon_{kl} \omega_l^a > 0 \\ \omega_i^a = 0 & \text{if } \epsilon_a = \omega_k^a \epsilon_{kl} \omega_l^a < 0 \end{cases} \quad (69)$$

In the first case, the strain normal to the crack is positive and the crack is assumed to be open; it contributes to the stiffness change by Equation (68). In the second case, with a negative normal strain, the crack is closed and inactive. The corresponding terms in Equation (68) are cancelled.

The theory is attractive in view of its simplicity to describe the unilateral conditions corresponding to opening/closure of the microcrack systems. In the macroscopic theories studied (Sections 3 and 4), the unilateral condition was written in a coordinate system associated with the principal directions of strains or stresses. Conversely here, the local coordinate system for the active/passive condition corresponds to the direction of the crack system.

However, this formulation is still unacceptable. Under a complex strain evolution, the stress response may show discontinuities associated with the closure of the microcracks. This deficiency can be illustrated by the following example.

In a two-dimensional case, let us assume the existence of two crack systems corresponding to the two directions $\vec{e}_1$ and $\vec{e}_2$ (Figure 3):

$$\vec{\omega}^1 = \omega_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \vec{\omega}^2 = \omega_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

When the two systems are active, and cracks open, the stiffness increment

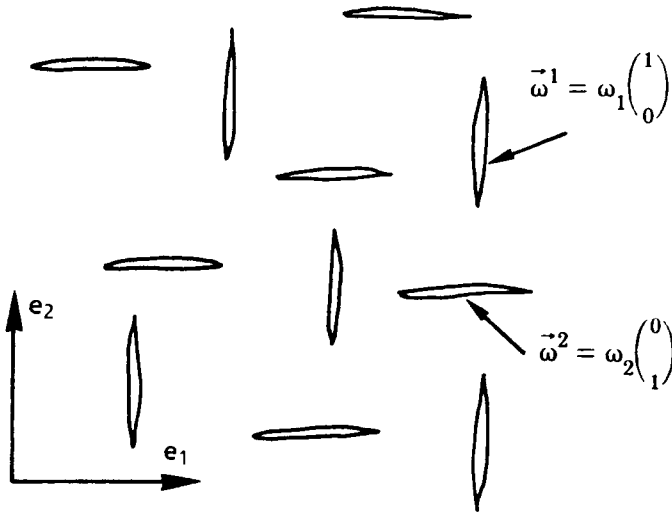


Figure 3. Example of bidirectional microcrack repartition (plane problem).

$\tilde{\Lambda}_{ijkl} - \Lambda_{ijkl}$ is obtained from Equation (68). In the system $(\vec{e}_1, \vec{e}_2)$, with the Voigt notations corresponding to $\epsilon_{11}, \epsilon_{22}, \epsilon_{12}$; it writes:

$$\Delta \Lambda = \begin{bmatrix} 2(C_1 + C_2)\omega_1^2 & C_1(\omega_1^2 + \omega_2^2) & 0 \\ C_1(\omega_1^2 + \omega_2^2) & 2(C_1 + C_2)\omega_2^2 & 0 \\ 0 & 0 & C_2(\omega_1^2 + \omega_2^2) \end{bmatrix} \quad (70)$$

Considering a strain evolution with $(\vec{e}_1, \vec{e}_2)$ being the principal strain directions, we can write the corresponding stress increment $\Delta \underline{\sigma} = (\tilde{\underline{\Lambda}} - \underline{\Lambda}) : \underline{\epsilon}$ for $\epsilon_1 > 0$ and $\epsilon_2 > 0$:

$$\begin{aligned} \Delta \sigma_1 &= 2(C_1 + C_2)\omega_1^2\epsilon_1 + C_1(\omega_1^2 + \omega_2^2)\epsilon_2 \\ \Delta \sigma_2 &= C_1(\omega_1^2 + \omega_2^2)\epsilon_1 + 2(C_1 + C_2)\omega_2^2\epsilon_2 \end{aligned} \quad (71)$$

If $\epsilon_1 < 0$ and $\epsilon_2 > 0$, the cracks of system (I) are closed, and from the condition (69), the stress increment is expressed by taking $\omega_1 = 0$:

$$\begin{aligned} \Delta \sigma_1 &= C_1\omega_2^2\epsilon_2 \\ \Delta \sigma_2 &= C_1\omega_2^2\epsilon_1 + 2(C_1 + C_2)\omega_2^2\epsilon_2 \end{aligned} \quad (72)$$

Assuming a (principal) strain evolution with $\epsilon_2 > 0$, the stress discontinuity

for σ_1 when ϵ_1 changes sign is obvious by comparing Equations (71) and (72) for $\epsilon_1 = 0$.

Let us insist on the discontinuous character. It is not a discontinuity in the rate of evolution of σ but in its actual value. For a continuous strain evolution, we obtain a sudden stress jump. Clearly such a behaviour is physically not acceptable.

5.2 The Theory of Andrieux et al. (1986)

In the framework of a theory where the slip and friction processes on the microcrack lips are also considered, which leads to some macroscopic plastic like dissipation, the authors introduce the unilateral condition of opening-closure of a microcrack system through a decomposition of the elastic compliance:

$$\begin{aligned} \tilde{\Lambda}^{-1} = \underline{\Lambda}^{-1} + \sum_a \frac{1}{K(\omega^a)} [\xi^a \tilde{n}^a \otimes \tilde{n}^a \otimes \tilde{n}^a \otimes \tilde{n}^a \\ + (\tilde{n}^a \otimes \tilde{t}^a)_s \otimes (\tilde{n}^a \otimes \tilde{t}^a)_s] \end{aligned} \quad (73)$$

$\tilde{n}^a$ is the unit normal to the microcrack system (parallel microcracks), and $\tilde{t}^a$ is the unit tangent to the crack. Here the formulation is written for plane problems and part-through cracks, where $\tilde{t}^a$ is completely defined. Generalization to more complex cases is possible without difficulty.

The key for the unilateral condition is the parameter ξ^a of each system, which takes the following values:

$$\xi^a = \begin{cases} 1 & \text{if } \sigma_a = \tilde{n}^a \cdot \underline{\sigma} \cdot \tilde{n}^a > 0 \\ 0 & \text{if } \sigma_a < 0 \end{cases} \quad (74)$$

Closure is assumed when the stress normal to the crack is negative.

To discuss the consequences of this choice, let us consider the particular case where the cracks are perpendicular to the direction $\tilde{e}_1$ of the coordinate system. In that case, the change of the compliance associated to the open microcracks does concern only two terms: Λ_{11}^{-1} , which corresponds to the normal stress component σ_{11} and Λ_{12}^{-1} which corresponds to the shear stress σ_{12} . When the crack closes, its influence on the first term of the compliance disappears. Andrieux et al. assume that when neglecting the friction on the crack lips, the shear resistance of the material is still affected by the crack.

Clearly the proposed formulation preserves the symmetry of compliance and the continuity of the strain evolution for any stress history. This property can be easily demonstrated from the fact that the only term affected by the unilateral condition is the form $\xi \tilde{n} \otimes \tilde{n} \otimes \tilde{n} \otimes \tilde{n}$. The corresponding strain can be written with indicial notations $\Delta \epsilon_{ij} = \xi n_i n_j n_k n_l \sigma_{kl} = \xi n_i n_j \sigma_a$. When ξ changes its value from 1 to 0, the normal stress σ_a is just zero, so that the strain response is not affected by the discontinuity of the compliance.

Such a formulation seems to be attractive, because it solves the problem of stress-strain discontinuities for the unilateral condition. However, it suffers two deficiencies:

- For the present writing, in terms of the compliance, it does not correctly allow the elastic behaviour of a solid with parallel cracks, as given by some quasi analytical solutions [see Delameter and Herman (1979) and Hoenig (1979)]. Moreover, the closure criterion can only be taken on the normal stress to each crack. This fact limits the applicability for brittle materials like concrete in which, as discussed by Simo and Ju (1987), the normal strain is more appropriate.
- By using the reverse method, writing an equation like Equation (73) for the elastic stiffness of damaged material, the closure criterion may be taken on the normal strain (to each crack), while still preserving the continuity of the stress-strain response. However, in that case, the described elastic behaviour would be significantly different from the one given by the above-mentioned exact solutions.

6. CONCLUSION

We have discussed the description of anisotropic damage in the framework of CDM theories based on the effective stress concept and the strain equivalence. Five typical theories have been selected that introduce the possibility for unilateral conditions for damage effect (active-passive) together with a differentiation of damage growth between tensile-like and compressive-like conditions.

The chosen theories are based on the following damage variables:

- two scalars, D , and D_c , in the theory developed by Ladeveze, Lemaitre (1984) and Mazars (1985)
- two second-order tensors $\underline{d}$ and $\underline{d}^*$ in the theory proposed by Ramtani (1990) generalizing the works of Cordebois and Sidoroff (1979)
- a fourth-order symmetric tensor $\underline{\underline{C}}$ in the theory developed by Ju (1989) with two different evolution rules for tensile and compressive conditions
- damage described by vector variables in the theories by Krajcinovic and Fonseka (1981) or by Marigo (1986)

These theories have been discussed in detail assuming plane stress (or plane strain) in the principal stress (strain) system. Several properties were considered for both the elastic behaviour and the damage loading surfaces:

- the capability to describe the anisotropic elastic behaviour induced by damage in an initially isotropic material
- the symmetry of the damaged elastic stiffness tensor
- the continuity of stress evolution (strain) when the unilateral condition for active/passive damage takes place
- the continuity and the convexity of the damage loading surface (in stress or strain spaces)

Table 1 summarizes the answers to these questions:

Table 1. The capabilities of the selected theories: (1) with scalars, (2) with a second-order tensor, (3) with a fourth-order tensor, (4) with vectors.

Theory	Complete Anisotropy	Symmetry of Elastic Stiffness	Continuity of Stress-Strain Response	Convexity of Loading Surface
1	No	Yes	Yes	No
2	Yes	No	Yes	Yes
3	No	Yes	No	Yes
4	Yes	Yes	No	Not studied

1. The theory with scalar variables is unable to reproduce anisotropy of damage. The symmetry of elastic behaviour and the continuity of stress-strain response are correctly obtained. However, the damage loading surface is not convex.
2. The theory with two second-order symmetric tensors and a scalar is the only one that reproduces anisotropy of damage after any kind of loading condition. The stress-strain response is continuous when the unilateral condition takes place. However, the elastic operator can be nonsymmetrical for complex loading conditions, which is not acceptable.
3. The theory using the present elastic stiffness tensor as a fourth-order symmetric damage variable describes an anisotropic behaviour for some loading conditions. However, the damage induced by positive principal strains is always isotropic. Moreover, the proposed active/passive unilateral damage condition leads to a discontinuous stress-strain response, which is not acceptable.
4. In the last section, we have also studied two theories based on a direct microcrack description as proposed, for instance, by Krajcinovic and Fonseka (1981) using a family of vectors to describe the damage. In fact, a discontinuity of the stress-strain evolution still appears with their formulation when the microcrack closure takes place! Another simple decomposition of the elastic compliance of a material with microcracks, chosen by Andrieux, Bamberger, and Marigo (1986), preserves the continuity, but the closure criterion must be limited to the normal stress to the crack. Its transposition in terms of the elastic stiffness and a normal strain criterion would reduce the correctness of the elastic response of the microcracked material.

From this study we conclude that presently existing CDM theories are able to describe the damage induced anisotropy or the unilateral active/passive damage conditions. However, there is no acceptable theory to simultaneously describe the two effects. The opinion of the author is that the lack of existing theories to describe both effects is general. Further research must be conducted in order to achieve the development of an acceptable theory in the framework of the macroscopic phenomenological approach to Continuum Damage Mechanics.

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